

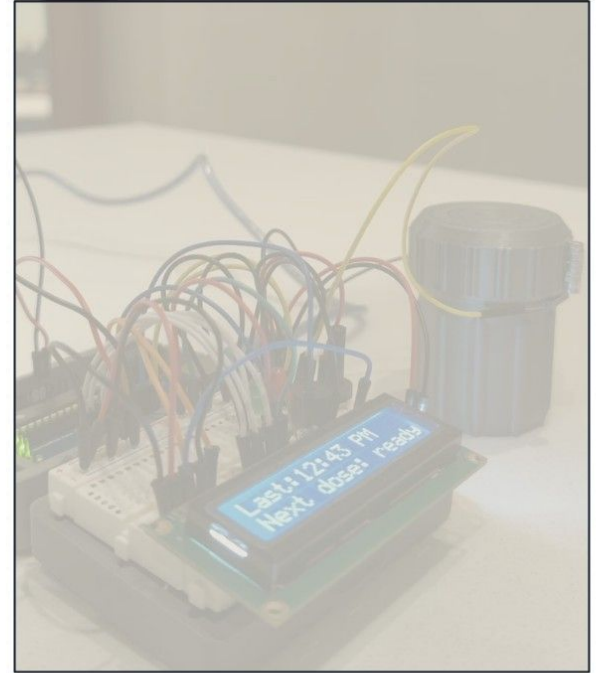
Safe|Snap

Snap It Shut. Stay Safe.

• SmartMedicationSafety



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THE PROBLEM & BACKGROUND:

Double Dosing: A Neglected Medication Crisis

125K

deaths/year from non-adherence (US)

50%

of chronic disease patients

\$300B

annual avoidable healthcare costs

How It Happens

The mechanism is almost always **uncertainty, not negligence.**

A patient takes a morning dose, becomes distracted, and cannot recall whether they took it. Faced with the choice of missing a dose or re-dosing. Many people may choose to re-dose

Vulnerable Populations

Elderly Patients managing 5+ medications with cognitive decline

Children at risk from caregiver coordination failures

Dementia Patients regardless of caregiver involvement

THE GAP:

Why Current Solutions Don't Solve Double Dosing

Pill Organizers

X Tell you if a compartment is empty, BUT not WHEN it was emptied or by WHOM

Phone Alarms

X Remind you to take a pill, BUT can't confirm whether you already did

Blister Packs

X Show consumed doses, BUT offer no real-time warning if you open one too soon

Smart Caps (Exist)

X Expensive (\$50–\$200+), subscription-based, not designed for all bottle types

OUR SOLUTION: What Is It?

The Product

SafeSnap is a smart pill bottle—not just a cap or attachment—that prevents accidental double dosing through passive magnetic cap detection and real-time dose logging.

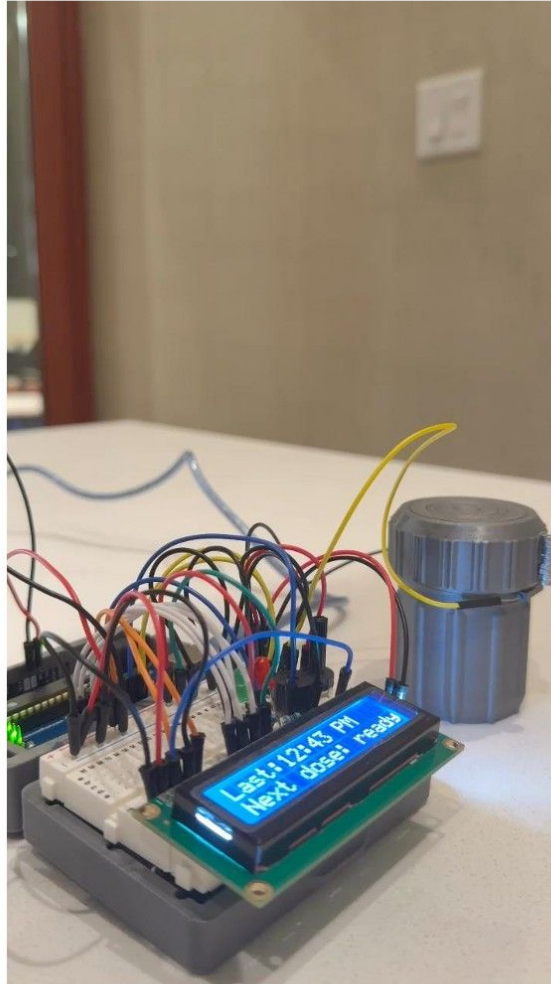
When you open the bottle, it detects it. When you close it, the timer starts. NO buttons, NO apps. NO behavior change required.

OUR SOLUTION: How It Works?

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1

Cap Screwed On

Magnet in cap aligns with reed switch. Circuit closed, system idle.

2

Cap Removed

Magnet moves away. Reed Switch Opens. Arduino logs the timestamp instantly.

3

Timer Checks Interval

Safe Window Elapsed → **Green LED**
Too Soon → **Red LED** + Buzzer Alert

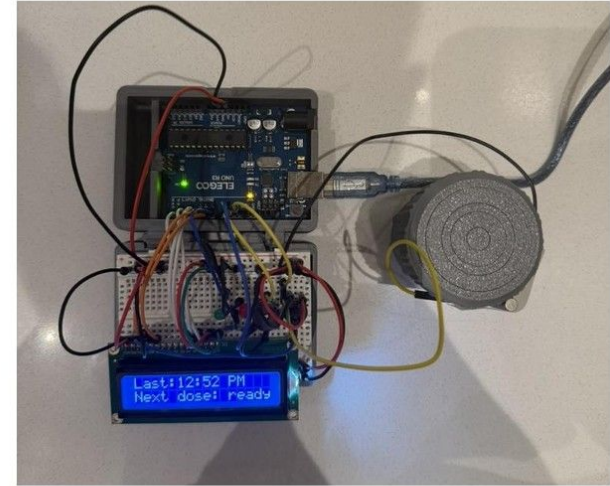
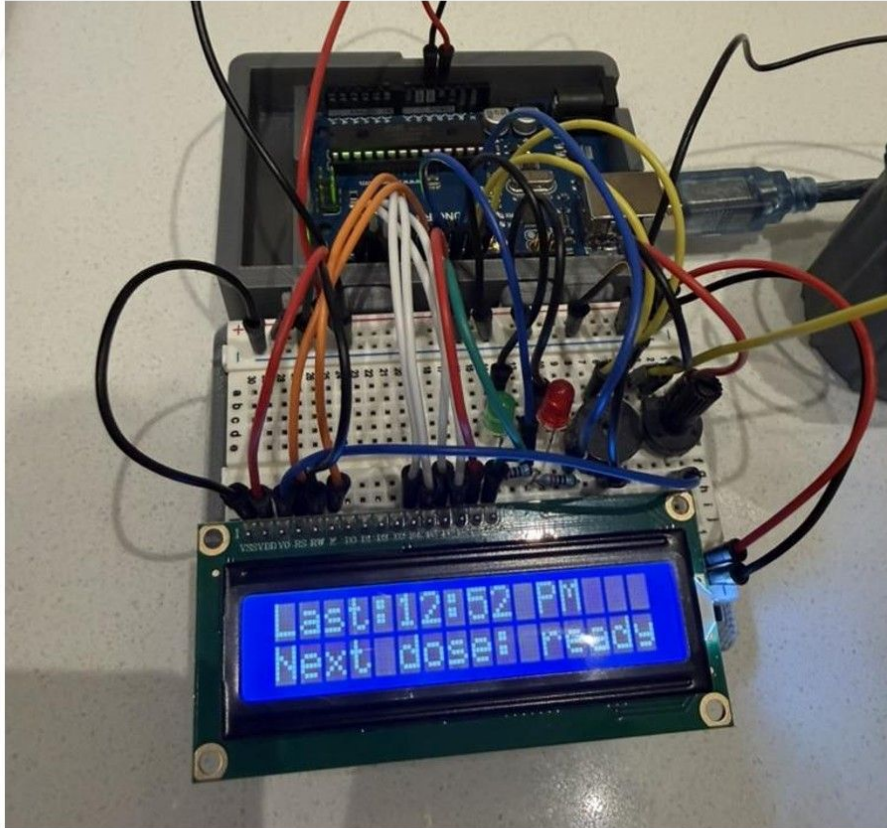
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LCD Confirms Status

“Last: 6:47 AM / Next dose: ready”
Always visible, no action needed.

OUR SOLUTION:

Working Prototype—Full Circuit & LCD Display



Arduino Uno R3

Main Controller—reads reed switch, drives LCD + alerts

LCD Screen

Displays last dose time and safe window in real time

Reed Switch + Magnet

Passive cap removal detection—no user action required

LED + Piezo Buzzer

Visual + audio alert for unsafe dose attempts

ITERATIVE DESIGN: From V1 to V2—Prototype Evolution

V1: Initial Design ✗



Cap did not close or screw tightly, threading flaws in CAD

V2: Redesigned ✓



Shorter, tighter threading, cap screws securely, reed switch aligns reliably

COST ANALYSIS: Accessible by Design?

Bill of Materials—Prototype Cost

Component	Item	Cost
Arduino Uno Kit	Microcontroller + Jumper Wires	Provided
Reed Switches	Plastic Reed Switch Pack	\$6.99
Piezo Buzzer	Disc Magnet Set	\$5.99
LED Assortment	Active Buzzer Pack	\$3.97
Neodymium Magnets	Red/Green Indicators	\$5.52
3D Printing	PLA/PETG Filament	Provided
TOTAL		\$~22.47

VS.

The Competition

Smart Cap (Hero Health)
~\$99 device + \$39.99/mo

Smart Cap (AdhereTech)
~\$150–\$200 device

Pill Organizer (Smart)
~\$35–\$80 device

SafeSnap Prototype
<\$30 all-in

SafeSnap At Scale
Est. \$10–\$15/unit

OUR SOLUTION:

What Makes SafeSnap Different?



Innovation

Passive reed switch detection—the act of opening the bottle IS the input. No buttons, no confirmation bias. The bottle itself is the product.



Healthcare Impact

Directly addresses double-dosing—a distinct medication error causing cardiovascular events, stimulant toxicity, and hospitalizations.



Functionality

Reed switch detects every cap removal. LCD shows dose time + countdown. LED + buzzer fire on unsafe intervals. Fully demonstrated live.



Usability

No app. No subscription. No buttons. Works with existing habits — open the bottle as normal. Designed for elderly, caregivers, parents.



Scalability

Next version integrates all hardware inside the bottle. BOM under \$30 now—manufacturable at scale for ~\$10–\$15/unit.



Cost-Effective

Built for <\$30 vs. \$50–\$200 existing smart caps. No subscription. One-time cost accessible to all income levels.

IMPACT & INNOVATION: FUTURE DEVELOPMENT

NOW

Working Prototype

- 3D-Printed Bottle + Cap (V2)
- External Breadboard + Arduino Uno
- LCD Screen: Last Dose + Next Window
- Reed Switch Detection—Live & Working
- LED + Buzzer Alert System

NEXT

Integrated Unit

- All electronics embedded inside bottle body
- LCD built into bottle exterior
- Battery-powered—No USB required
- Wire channels in CAD Design
- Single self-contained product

FUTURE

Full Product

- Bluetooth/Wi-Fi caregiver alerts
- Companion mobile app
- Multi-medication tracking
- Clinical validation study
- Pharmacy partnership distribution
- Sensor Fusion

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Questions?

